Object Detection for Autonomous Driving

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Abstract

This paper explores the use of YOLOv5 and YOLOv8 models for vehicle object detection. Two different datasets, including an augmented dataset, were used to train and evaluate the models. The performance of the models was assessed based on precision, recall, and Mean Average Precision (mAP). The results showed that YOLOv8 trained on the augmented dataset achieved the highest mAP score. The study also identified challenges in detecting vehicles near image edges. Future work may involve addressing this limitation and further improving the models' performance.

1. Introduction

Autonomous driving cars have increased their popularity in the past years due to its benefits and applications. Between them, can be mention accident prevention, since the vehicle is capable of measuring and estimating parameters of the environment, such as other cars, pedestrians, etc. hence improving safety while driving.

Furthermore, now a days cars, trucks and other means of transport have become a necessity therefore increasing its density in the highways or streets which augment traffic congestion, however, with the usage of autonomous vehicles an optimization of highways is becoming achievable thanks to the development of technologies for addressing congestion which aim to generate "traffic friendly" adaptive cruise control systems, among many more benefits autonomous vehicles bring.

Moreover, to create these technologies is necessary to develop a precise object detection software which is able to differentiate components of the driving scene while being moving, in order to provide with information to the vehicle and help it decide what action to take. To develop the required software, Neural Networks are trained, which are algorithms designed to recognize patterns, learn from data, and make predictions or decisions.

For better convenience, prebuilt object detection models can be utilized, such as YOLO (You Only Look Once) models, which has gained fame in the recent years since it can deliver fast and accurate real-time results.

2. Related Work

Recent research in vehicle object detection has explored different approaches to improve accuracy and efficiency. Prominent methods include region-based convolutional neural networks (R-CNN) like Faster R-CNN [4] and single-shot detection (SSD) models like YOLO.

Efforts have been made to enhance detection accuracy through ensemble models, feature fusion, and attention mechanisms. Data augmentation techniques have also been utilized to improve model generalization.

As part of the literature employed for the development of the project, official YOLOv5[2] and YOLOv8[3] documentation was used as a guide to understand how to implement the models.

3. Methods

In this section, an overview of the two object detection models used in this study is presented.

3.1. YOLOv5

YOLOv5 [2], short for "You Only Look Once v5," is a state-of-the-art real-time object detection model. It builds upon the previous versions of YOLO and introduces several architectural improvements for enhanced performance.

The YOLOv5 architecture consists of a backbone network, which extracts features from the input image, followed by detection heads that perform object detection at multiple scales. YOLOv5 uses a combination of convolutional layers, including residual blocks and focus modules, to capture rich spatial and semantic information.

One key characteristic of YOLOv5 is its focus on model efficiency. It strikes a balance between accuracy and computational efficiency by employing a "Scaled-YOLO" approach, which allows for flexible model size choices during training and inference.

3.2. YOLOv8

YOLOv8 [3], also known as "You Only Look Once v8," is an evolution of the YOLO architecture that further improves object detection performance. It incorporates advancements in network design, feature extraction, and prediction techniques.

Similar to YOLOv5, YOLOv8 follows a single-shot detection paradigm, where object detection is performed in a single pass over the input image. It employs a deep neural network with a backbone architecture, which extracts hierarchical features from the input.

In this study, both YOLOv5 and YOLOv8 were trained and evaluated for vehicle object detection using custom datasets. The models were trained on GPUs and underwent experimentation to optimize their performance. The resulting models were evaluated based on precision, recall, and Mean Average Precision (mAP) metrics, providing insights into their effectiveness for vehicle detection tasks.

4. Evaluation

4.1. Experimental Setup

4.1.1 Dataset

Provided data for training and evaluating the models is a reduced and downscaled version of the KITTI [1] dataset, an example is shown in the Figure 1. The KITTI dataset is a widely used benchmark dataset in the field of autonomous driving and computer vision, providing labeled data for various tasks, including object detection.

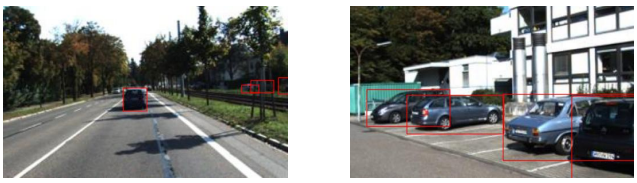
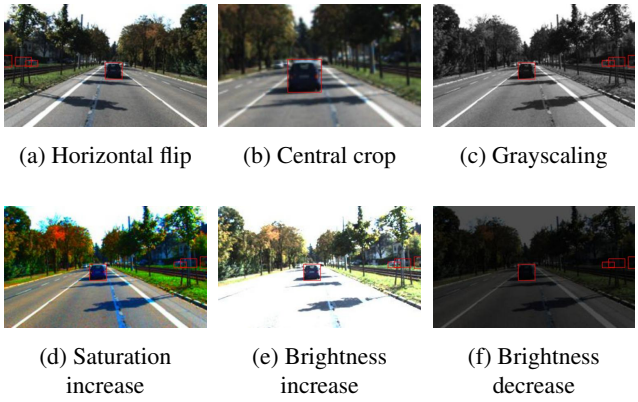


Figure 1: Two images with bounding boxes from the KITTI [1] dataset

The dataset was divided into two distinct subsets: a training set and a test test. While the training set was labeled, the test set was purposely unlabeled, introducing a challenge to evaluate the models' performance on unseen data. To overcome this challenge and estimate the networks' generalization ability, the training set was further partitioned into a separate train and validation sets.

Prior to training, all images underwent resizing. The original images, initially sized at 306×185 pixels, were uniformly resize to dimensions of 320×193 pixels. This resizing process was crucial to meet the requirement that the image size be a multiple of the models' maximum stride, which is 32.

To further enhance the generalization and robustness of the models, five image augmentation techniques were employed. Firstly images from the training and validation sets were converted to grayscale (Figure 3c), given a higher saturation value (Figure 3g) and given both a higher and lower brightness value (Figures 3h and 3i). Finally, all resulting images (including the original image) were flipped horizontally (Figure 3a) and cropped in the center (Figure 3b). By applying these augmentation techniques, 20 unique images were generated from a single original image. The number of images for each original and augmented sets stated in Table 1.



(a) Horizontal flip (b) Central crop (c) Grayscaling

(d) Saturation increase (e) Brightness increase (f) Brightness decrease

Figure 2: Augmented images

Dataset	Original	Augmented
Train	2 969	59 380
Val	743	14 860
Test	3 769	-

Table 1: Number of images in training, validation and test sets of the original and augmented datasets

4.1.2 Hardware & software

To perform the object detection task using YOLOv5 and YOLOv8 models trained on a custom dataset, we utilized specialized hardware and software tools. The hardware setup involved the use of the NVIDIA DGX-2[1], a powerful deep learning workstation provided by the University of Lübeck. The DGX-2 workstation is equipped with high-performance components specifically designed for accelerated deep learning tasks. The NVIDIA DGX-2 16 NVIDIA® Tesla® V100 GPUs, which significantly expedite the training process by parallelizing computations. In

[1] https://www.nvidia.com/en-gb/data-center/dgx-2/

our experiments, we utilized four GPUs to leverage their computational power for training the models. This setup allowed us to distribute the workload across multiple GPUs, resulting in faster training times and improved efficiency.

As a programming language Python 3.11.3[2] was chosen, as it is a widely used language in the field of machine learning and computer vision. To augment the dataset and perform data preprocessing PyTorch 2.0[3] library was utilized. PyTorch provided a comprehensive set of tools for image augmentation, enabling to apply transformations such as rotation, scaling, and cropping to enhance the diversity and variability of the data. Additionally, PyTorch and Ultralytics[4] libraries were used for loading of the YOLOv5 and YOLOv8 models.

4.1.3 Training

To train YOLOv5 and YOLOv8 models for vehicle object detection two different datasets were used: the original dataset and an augmented dataset. As training process was performed utilizing four GPUs this enabled concurrent computation and expedited the training time. Taking advantage of the available computational resources, a batch size of 64 images was selected for each iteration. Each YOLOv5 and YOLOv8 model underwent training for a total of 500 epochs. The chosen number of epochs allowed the models to converge towards optimal performance and achieve robustness in vehicle object detection.

In total four model were trained:

- YOLOv5 on the original dataset (YOLOv5 ORIG)
- YOLOv5 on the augmented dataset (YOLOv5 AUG)
- YOLOv8 on the original dataset (YOLOv8 ORIG)
- YOLOv8 on the augmented dataset (YOLOv8 AUG)

4.2. Discussion

From the evaluation results shown in Table 2, it can be observed that the YOLOv8 AUG model, trained and eveluated on the augmented dataset, achieved the highest mAP_{50} score of 0.983. This outcome aligns with the expectations, as YOLOv8 is a newer version of the YOLO architecture, which often exhibits improved performance. The use of augmented data in the training process contributed to enhancing the detection capabilities of the YOLOv8 model, resulting in a higher mAP_{50} score.

Conversely, the YOLOv5 model trained with the original dataset and evaluated on the augmented data obtained the lowest mAP_{50} score among the models. This indicates that the YOLOv5 model's performance was not as robust when faced with the augmented dataset, potentially due to the limited diversity and variability in the original training data.

Model	Dataset	Precision	Recall	mAP_{50}
YOLOv5 ORIG	ORIG	0.964	0.932	0.974
	AUG	0.919	0.911	0.950
YOLOv5 AUG	ORIG	0.981	0.941	0.973
	AUG	0.976	0.939	0.976
YOLOv8 ORIG	ORIG	0.974	0.943	0.979
	AUG	0.910	0.892	0.951
YOLOv8 AUG	ORIG	0.976	0.958	0.981
	AUG	0.980	0.953	0.983

Table 2: Evaluation of the val dataset

Model	Average confidence score
YOLOv5 ORIG	0.8776
YOLOv5 AUG	0.8785
YOLOv8 ORIG	0.8671
YOLOv8 AUG	0.8715

Table 3: Predictions on the test dataset

Nevertheless, it is important to note that all models achieved acceptable results, as their mAP_{50} values were equal to or greater than 0.95. These scores indicate that the models were effective in accurately detecting vehicles, even though with varying degrees of precision and recall.

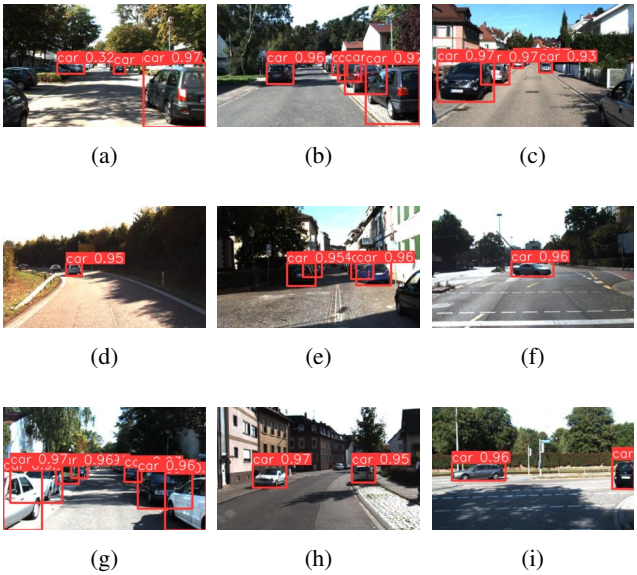
Table 3 presents the average confidence scores of the models, representing the models' level of certainty in their predictions. The scores provide insights into the models' confidence in detecting vehicles in the test dataset.

Upon examination of the average confidence scores, it can be observed that there are no significant differences among the models. All models exhibited similar average confidence scores, ranging from 0.8671 to 0.8785. This suggests that the models displayed comparable levels of confidence in their predictions, regardless of the dataset or architecture used.

To provide a more comprehensive understanding of the models' performance, we present visual representations of the detected objects in Figure 3. These images showcase the models' ability to accurately identify and localize vehicles in various scenarios.

Despite the overall satisfactory performance of the trained models, it is important to acknowledge their limitations and challenges encountered during the evaluation

[2]https://www.python.org
[3]https://pytorch.org
[4]https://ultralytics.com



(a) (b) (c)

(d) (e) (f)

(g) (h) (i)

Figure 3: Prediction results with bounding boxes

process. One notable limitation is the difficulty faced by the models in detecting vehicles near the edges of the images.

The issue of edge detection can be addressed through various strategies, such as utilizing larger image sizes during training, applying padding techniques to expand the field of view, or implementing post-processing methods to refine the detection results near the image boundaries. These approaches could potentially enhance the models' ability to detect vehicles in challenging scenarios and improve their overall performance.

5. Conclusion and Future Work

YOLOv5 and YOLOv8 models were trained and evaluated for vehicle object detection using both original and augmented datasets. The models were trained on a custom dataset derived from the KITTI dataset, and their performance was assessed based on precision, recall, and Mean Average Precision at IoU threshold 0.50 (mAP_{50}) metrics.

The evaluation results demonstrated that the YOLOv8 model trained on the augmented dataset achieved the highest mAP_{50} score of 0.983, highlighting its superior performance compared to the other models. The use of augmented data proved beneficial in enhancing the detection capabilities of the YOLOv8 model. However, all models exhibited acceptable performance, with mAP_{50} values greater than or equal to 0.95, indicating their effectiveness in vehicle object detection.

While the models showed promising results, a limitation was observed in their ability to detect vehicles near the image edges. This issue calls for further investigation and improvement to enhance the models' performance in challenging scenarios. The performance of the models could be enhanced by incorporating additional data augmentation techniques specifically designed to simulate edge cases, such as vehicles partially occluded or located at the edges of the images.

References

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